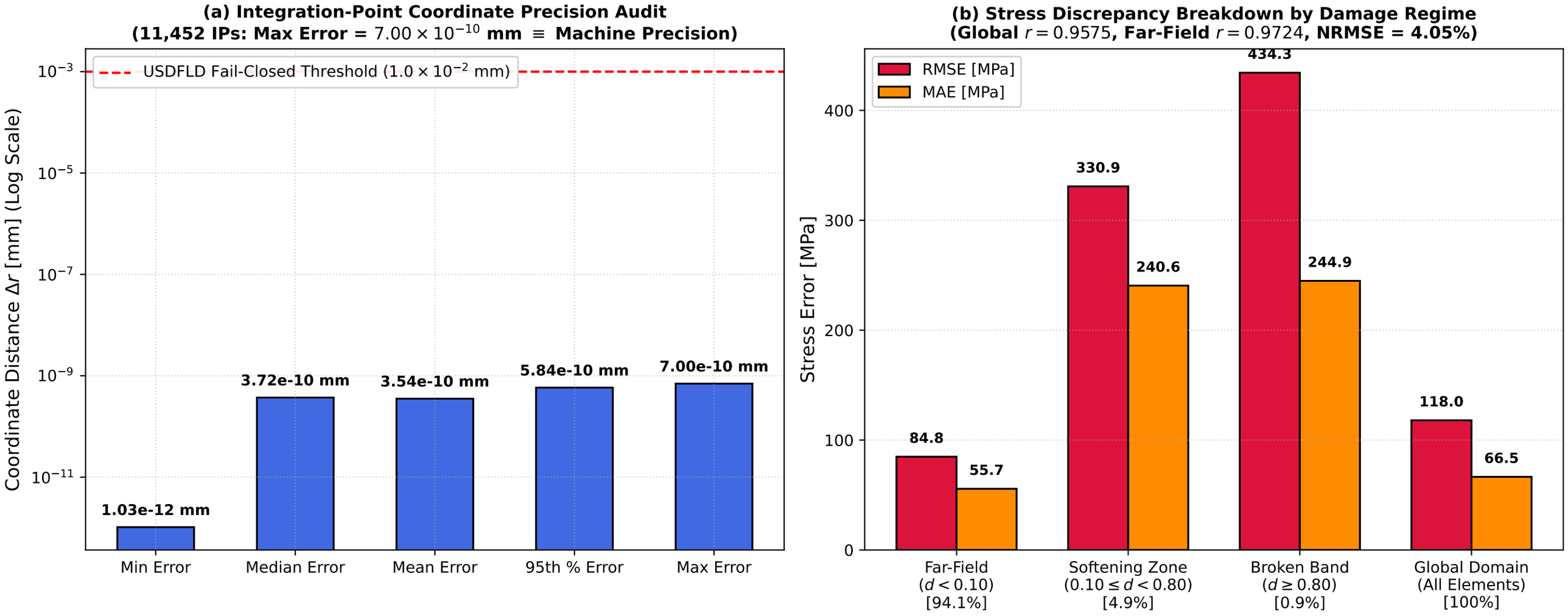
(a) Integration-Point Coordinate Precision Audit
(11,452 IPs: Max Error = 7.00×10^{-10} mm ≡ Machine Precision)
USDFLD Fail-Closed Threshold (1.0×10^{-2} mm)
Coordinate Distance Δr [mm] (Log Scale)
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
1.03e-12 mm
3.72e-10 mm
3.54e-10 mm
5.84e-10 mm
7.00e-10 mm
Min Error
Median Error
Mean Error
95th % Error
Max Error
(b) Stress Discrepancy Breakdown by Damage Regime
(Global $r = 0.9575$, Far-Field $r = 0.9724$, NRMSE = 4.05%)
RMSE [MPa]
MAE [MPa]
Stress Error [MPa]
0
100
200
300
400
84.8
55.7
330.9
240.6
434.3
244.9
118.0
66.5
Far-Field
($d < 0.10$)
[94.1%]
Softening Zone
($0.10 \leq d < 0.80$)
[4.9%]
Broken Band
($d \geq 0.80$)
[0.9%]
Global Domain
(All Elements)
[100%]